Worksheet 10 - Product and Quotient Rule, Trig Derivatives

MATH 1271

10/13/2026

1. Use relevant derivative rules to differentiate each function below.

   (a) $h(z) = \sqrt{z} + \dfrac{1}{z^4} + 5^z$ (Please use Leibniz notation.)

   (b) $r(x) = \dfrac{3^x}{x^4 + 1}$

   (c) $p(t) = (t^3 - t^2 + t - 1)(t^2 + 3t + 1)$

   (d) $g(y) = \dfrac{y^2 e^y}{2y + 1}$

2. Suppose that f and g are differentiable functions such that

$$f(4) = 2, \qquad f'(4) = 6, \qquad g(4) = 5, \qquad g'(4) = -3$$

Using this information, compute the derivatives of each of the following functions:

(a) $h(x) = f(x)g(x)$. What is $h'(4)$?

(b) $S(t) = \dfrac{f(t)}{g(t)}$. What is $S'(4)$?

(c) $q(z) = z^2 f(z) + z$. What is $q'(4)$?

(d) $k(x) = \dfrac{x}{f(x)}$. What is $k'(4)$?

3. Suppose that a bakery sells cakes. Each cake is priced at $p = 20$ dollars. At this price, the company sells 500 cakes per month and demand is falling at a rate of 30 cakes (sold per month) per dollar.

   The bakery is looking over its finances. In particular, it is looking at its *monthly revenue*:

$$\text{Monthly Revenue} = \Big(\text{Price per cake}\Big) \times \Big(\text{Number of cakes sold per month}\Big)$$

   The bakery would like to increase its monthly revenue. To do this, the bakery is considering one of the following options:

   - increase the price of a cake to $p = 21$ dollars;
   - keep the price of a cake at $p = 20$ dollars;
   - decrease the price of a cake to $p = 19$ dollars.

   In this problem, we'll use calculus to help us determine what the bakery should do. Let $f(p)$ be the number of cakes sold (per month) when the price of a cake is set to p dollars.

   (a) How can the statement "when each cake is priced at $p = 20$ dollars, the bakery sells 500 cakes per month" be rephrased in terms of the function $f(p)$ and/or its derivative, $f'(p)$?

   (b) How can the statement "when each cake is priced at $p = 20$ dollars, demand is falling at a rate of 30 cakes per dollar" be rephrased in terms of the function $f(p)$ and/or its derivative, $f'(p)$?

   (c) Let $R(p)$ be the monthly revenue (in dollars) that the bakery earns each month from the sale of cakes. What is a formula for $R(p)$ in terms of p and $f(p)$?

(d) Use the Product Rule to find a formula for $R'(p)$ in terms of p, $f(p)$, and $f'(p)$.

(e) At what rate is the monthly revenue changing when the price of a cake is set to $p = 20$?

(f) Based on your answer to the previous part, what should the company do? Should it raise the price of a cake? Keep it the same? Or decrease the price of a cake? What would lead to higher monthly revenue? Why?

4. Calculate each of the following derivatives:

(a) $f(x) = (5x^3 + \sin(x))(4^x - 2\cos(x))$

(b) $p(t) = \dfrac{\cos(t)}{t^6 \cdot 6^t}$

(c) $g(r) = \dfrac{r\sec(r)}{5^r}$

5. Use the derivative rules that have been covered so far to answer each of the following questions about slope.

   (a) Find the (exact) slope of the tangent line to the graph of $y = 2x + \dfrac{\sin(x)}{2}$ at the point where $x = \frac{\pi}{6}$.

   (b) Find the (exact) slope of the tangent line to the graph of $y = x^2 \cos(x)$ at the point where $x = \frac{\pi}{2}$.

(c) Let $f(\theta) = \theta - 2\sin(\theta)$. On the interval $[0, 2\pi]$, at which value(s) of θ does the graph of $f(\theta)$ have a horizontal tangent line?

(d) Let

$$g(x) = \tan(x) + x, \qquad 0 \leq x \leq 2\pi.$$

At which values of x (between 0 and 2π) do we have $g'(x) = 3$?

6. The function

$$P(t) = 24 + 8\sin(t)$$

represents a population of a particular kind of animal that lives on a small island. Here, P is measured in hundreds and t is measured in decades since January 1, 2010.

(a) What is the instantaneous rate of change of P on January 1, 2030? (Round to the nearest 0.01 and include units on your answer.)

(b) Write a sentence in everyday language that explains how the population is behaving at this point in time.

(c) Was the population increasing or decreasing in size on January 1, 2025? At what rate?

(d) (A warm up for future classes:) Between January 1, 2010 and January 1, 2050, on what date (month and year) do we expect the population to be at its peak (as large as possible)?

Hint: Graph $P(t)$ on the interval $[0, 4]$. Notice that at the peak population, the tangent line is horizontal (slope 0). At what value of t will this happen? How do you translate this value of t to a month and year?